

product_name: Food Delivery App

product_summary: A mobile application that enables customers to browse local restaurants, place food orders, track their delivery in real-time, and provide feedback on their experience.

problem_statement: Customers often face inconvenience when ordering food, including limited options, lack of real-time tracking, and difficulty providing feedback, leading to frustration and uncertainty about their orders.

target_users: ['Customers', 'Restaurants (Owners/Managers)', 'Delivery Drivers', 'Administrators']

core_features: ['Customer Account Management', 'Restaurant Browsing and Search', 'Menu Viewing and Item Selection', 'Order Placement and Customization', 'Payment Processing', 'Real-time Order Tracking', 'Order History', 'Rating and Review System', 'Restaurant Management (for restaurants)', 'Driver Management (for drivers)', 'Delivery Assignment and Management', 'Notification System']

mvp_features: ['Customer Registration/Login', 'Browse Restaurants and Menus', 'Add items to cart', 'Place Order', 'Basic Order Tracking (status updates)', 'Payment Integration (single method)', 'Restaurant Order Reception', 'Basic Delivery Driver Assignment', 'Customer Feedback submission']

modules: ['Customer Frontend (Mobile App)', 'Restaurant Partner Portal (Web/Mobile)', 'Delivery Driver App (Mobile)', 'Backend Services (API Gateway, Order Management, User Management, Restaurant Management, Delivery Management, Payment Gateway Integration, Notification Service)', 'Admin Dashboard']

data_entities: ['User (Customer, Restaurant, Driver, Admin)', 'Restaurant', 'Menu Item', 'Order', 'Order Item', 'Delivery', 'Payment Transaction', 'Review/Rating', 'Address', 'Notification']

possible_apis: ['Payment Gateway API (e.g., Stripe, PayPal)', 'Mapping/Location Services API (e.g., Google Maps, Mapbox)', 'SMS/Email Notification API (e.g., Twilio, SendGrid)', 'Push Notification Service API (e.g., Firebase Cloud Messaging)']

functional_requirements: ['FR1: The system shall allow customers to register and log in securely.', 'FR2: The system shall display a list of available restaurants based on the customer's location.', 'FR3: The system shall allow customers to browse restaurant menus and view item details.', 'FR4: The system shall enable customers to add/remove items to/from a shopping cart.', 'FR5: The system shall allow customers to place an order, specifying delivery address and payment method.', 'FR6: The system shall process payments securely.', 'FR7: The system shall provide real-time updates on order status (e.g., "Order Placed", "Preparing", "Out for Delivery", "Delivered").', 'FR8: The system shall allow customers to track the delivery driver's location on a map.', 'FR9: The system shall enable customers to submit ratings and reviews for restaurants and delivery experiences.', 'FR10: The system shall allow restaurants to receive, accept, and manage incoming orders.', 'FR11: The system shall allow delivery drivers to accept delivery requests and update delivery status.', 'FR12: The system shall send notifications to customers, restaurants, and drivers regarding order updates.']

non_functional_requirements: ['NFR1: The application shall be available 99.9% of the time.', 'NFR2: The system shall process customer orders within 5 seconds.', 'NFR3: The system shall handle a minimum of 1000 concurrent orders.', 'NFR4: All sensitive customer and payment data shall be encrypted both in transit and at rest.', 'NFR5: The user interface shall be intuitive and easy to navigate for all user roles.', 'NFR6: The mobile applications shall be compatible with iOS and Android platforms.', 'NFR7: The system shall be scalable to accommodate future growth in users and orders.']

constraints: ['C1: Initial launch limited to a specific geographic region/city.', 'C2: Must integrate with at least one major payment gateway.', 'C3: Development timeline for MVP is 3 months.', 'C4: Budget limitations for third-party API usage.']

assumptions: ['A1: Customers have access to smartphones and internet connectivity.', 'A2: Restaurants have internet access and a device to manage orders.', 'A3: Delivery drivers own smartphones and have data plans.', 'A4: Legal and regulatory requirements for food delivery services are understood and will be adhered to.', 'A5: A sufficient number of restaurants and delivery drivers will onboard to the platform.']